Davin Win Kyi

Curriculum Vitae
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Seattle, WA

Research Interests

Augmented Reality, Intelligent User Interfaces, Human–Computer Interaction (HCI), Human–AI Interaction and Accessibility

Education

University of Washington — Seattle, WA
M.S. in Computer Science, GPA: 3.84/4.00 — Mar. 2024 –Dec. 2026 (Expected)

University of Washington — Seattle, WA
B.S. in Computer Science, GPA: 3.89/4.00 — Sep. 2021–Mar. 2024

Research Experience

Dub Group — Seattle, WA
Research Assistant — Sep. 2024–Present

- Brainstormed, built and deployed a user study on a system that assist users, such as users with cognitive accessibility needs, with progress-based tasks, which can be used on different webpages while supporting users' needs into these interfaces.
- Collaborated on tools that help software developers audit accessibility barriers in the applications that they create

Makeability Lab — Seattle, WA
Research Assistant — Nov. 2022–Present

- Collaborated with Microsoft Research (MSR) and Andy Wilson on a project that enables front viewing phone cameras to retrieve context into what user is viewing through the detection of objects in the corneal scan of a users pupil
- Built and evaluated 2 AR personalization systems: SonoCraftAR, an AR system that enables deaf and hard of hearing individuals to author real-time sound-reactive UIs that best fit their needs and PreviewAR, an AR system that allows users to retrieve products that meet their spatial and semantic preferences.
- Collaborated on a mobile AR translation system that translates surrounding text to a target language, while preserving its visual style

Human Centered Robotics Lab — Seattle, WA
Research Assistant — Nov. 2024–Mar 2025

- Built and tested a simulation through the use of IsaacLab and RCareWorld for an attachable mobile phone holder robot for users with mobility impairments to view their phone in varying positions

Robot Learning Lab — Seattle, WA
Research Assistant — Nov. 2024–Mar 2025

- Built an object dataset and trained 10+ robotic manipulation environments for tasks involving these objects with stable-state start positions to goal positions in Isaac Sim

– Collaborated in the development and training of fine-grained manufacturing tasks, such as screwing in a bolt into a nut, in simulation through Isaac Sim

University of Washington Seattle, WA
REU Research intern Sep. 2023–Dec. 2023

– Built and evaluated an AR system that assists blind and low vision users in crosswalk veering through the integration of real-time object detection, vision-language reasoning, and auditory guidance for 20 participants in various crosswalk settings

Professional Experience

Amazon Seattle, WA
Software Development Intern Jun 2024, Sep 2024

– Designed an LLM-assisted application on Amazon Bedrock and prototyped full-stack features enabling developers to generate, review, and refine Amazon.com design variations through human in the loop feedback which supported evaluation of thousands of variants and contributed to click-through-rate improvements.

Capital One San Francisco, CA
Software Engineering Intern Jun 2023–Aug 2023

– Built and deployed a Python and FastAPI application serving Capital One customers and supporting millions of requests annually by integrated internal APIs, containerized the application with Docker, and deployed it on Kubernetes, reducing processing time by 73%.

Ansys Canonsburg, PA
Software Engineering Intern Jan. 2023–Mar. 2023

– Improved Ansys Discovery's development and testing infrastructure by engineering 25+ batch scripts that reduced daily installation runtime by 20% for 20+ software engineers, adding 50+ C# tests in collaboration with three developers which reduced bugs by 15%, and resolving 100+ critical test failures which reduced bugs by 10%.

PennyMac Agoura Hills, CA
Software Engineering Intern Jun. 2022–Aug. 2022

– Developed customer acquisition and internal data-access tools using Salesforce Marketing Cloud, AMPscript, HTML, CSS, and JavaScript, partnered with the product team to automate campaign workflows, increasing lead acquisition by 20%, improving customer retention, and converting thousands of prospective leads, while building a secure login system used by hundreds of PennyMac employees.

Conference and Journal Papers

C.4 Mingyuan Zhong, Mengqi Shi, **Davin Win Kyi**, Ajit Mallavarapu, Xia Chen, James Fogarty, and Jacob O. Wobbrock. 2026. *InterA11y: Interactive Web Accessibility Error Detection with LLM Agents.* Submitted to ACM The Web Conference 2027 (WWW 2027). **In submission.** PDF

C.3 Jaewook Lee, Judith Amores, Bo-Eun Choi, Donghoon Shin, Jason Kim, **Davin Win Kyi**, Seok-Young Kim, Jon E Froehlich, Andrew D Wilson. *EyeQ: Exploring Corneal Reflection for Mobile Context Sensing.* Submitted to the 2027 CHI Conference on Human Factors in Computing Systems (CHI 2027). **In submission.**

C.2 Caleb Chang, **Davin Win Kyi**, Natasha Jaques, and Karen Leung. 2026. *Do as the Romans Do: Learning Universal Behaviors from Heterogeneous Agents.* Submitted to the 2027 International Conference on Machine Learning (ICML 2027). **In submission.** arXiv:2606.18537

C.1 Mingyuan Zhong, Xia Chen, **Davin Win Kyi**, Chen Li, James Fogarty, Jacob O Wobbrock. 2026. *TaskAudit: Detecting Functiona11ity Errors in Mobile Apps via Agentic Task Execution.* In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems* https://dl.acm.org/doi/full/10.1145/3772318.3791415

Peer-Reviewed Demos, Workshops, and Posters

P.2 Mingyuan Zhong, Ajit Mallavarapu, Mengqi Shi, **Davin Win Kyi** James Fogarty, and Jacob O. Wobbrock. 2026. *Targeted Interaction Agents for Web Accessibility Evaluation.* UIST Adjunct '26 TBD

P.1 **Davin Win Kyi**, Jason Kim, Caleb Hu, Jewoo Park, Jaewook Lee, and Jon E. Froehlich. 2026. *PreviewAR: Sketching Spaces for Scale-Aware Product Discovery in AR.* UIST Adjunct '26 https://dl.acm.org/doi/full/10.1145/3830397.3841872

D.1 Minbeom Kim, Jason Kim, **Davin Win Kyi**, Seok-Young Kim, Jaewook Lee, and Jon E. Froehlich. 2026. *TranslateAR: Towards Style-Preserving In-Situ Text Translation in Augmented Reality.* UIST Adjunct '26. https://dl.acm.org/doi/full/10.1145/3830397.3841898

W.1 Jaewook Lee, **Davin Win Kyi**, Leejun Kim, Jenny Peng, Gagyeom Lim, Jeremy Zhengqi Huang, Dhruv Jain, and Jon E. Froehlich. 2025. *SonoCraftAR: Towards Supporting Personalized Authoring of Sound-Reactive AR Interfaces by Deaf and Hard of Hearing Users.* IDEATExR Workshop, IEEE ISMAR 2025 (ISMAR-Adjunct). https://doi.org/10.1109/ISMAR-Adjunct68609.2025.00108

Teaching

University of Washington Seattle, WA
Teaching Assistant, Artificial Intelligence Sep. 2025–Jun. 2026

- Mentored and held office hours on artificial intelligence for topics ranging from Markov Decision Processes to Reinforcement Learning

University of Washington Seattle, WA
Teaching Assistant, Interaction Programming Jan. 2023–Jun. 2024

- Prepared and presented 20+ class sections with interactive coding demos, presentations, and exercises, developed 20+ course projects and materials for 300+ students in Android development using Java and guided 300+ students through Java, Android Studio, and XML over 50+ office hours and 20+ sections

University of Washington Seattle, WA
Teaching Assistant, Introduction into Databases Sep. 2022–Dec. 2022

- Guided 200+ students through database fundamentals through interactive presentations and coding demonstrations over sections and office hours for topics ranging from database design to transactions

Leadership & Community Engagement

University of Washington Seattle, WA
Burma Myanmar Student Association Tech Lead Sep. 2022–Jun 2024

- Supporting the facilitation of club events, activities and Burmese culture through website development and maintenance.

University of Washington Seattle, WA
Stealth Start Up Founder Mon. 2025–Present

– Facilitating start up meetings, logistics, and communication, leading front-end team development goals and facilitating outreach across the University of Washington.

Mentorship

- **Jewoo Park** Sep. 2025 – Present
  CSE Undergraduate, University of Washington
- **Caleb Hu** Sep. 2025 – Present
  CSE Undergraduate, University of Washington

Awards and Honors

Dean's List Award 2021-2024

References

Jon E. Froehlich
Professor, University of Washington